Grokking the Coding Interview: Patterns for Coding Questions

Fast track your coding interview prep by learning coding patterns. Act smartly, and follow the coding patterns.

Level: Intermediate | Study Time: 321h | Lessons: 480 | Playgrounds : 517

2,11,991 (Students)

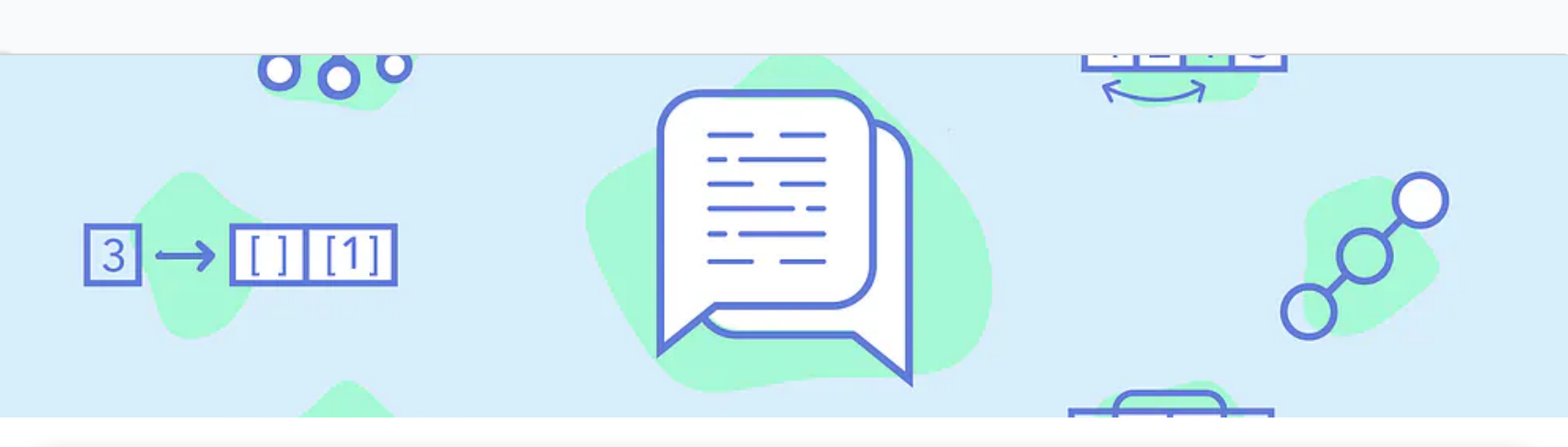
Course Overview

Coding interviews are getting harder every day. A few years back, brushing up on key data structures and going through 50-75 coding interview questions was more than enough prep for an interview. Today, everyone has access to massive sets of coding problems, and they have gotten more difficult to account for that. The process has gotten more competitive.

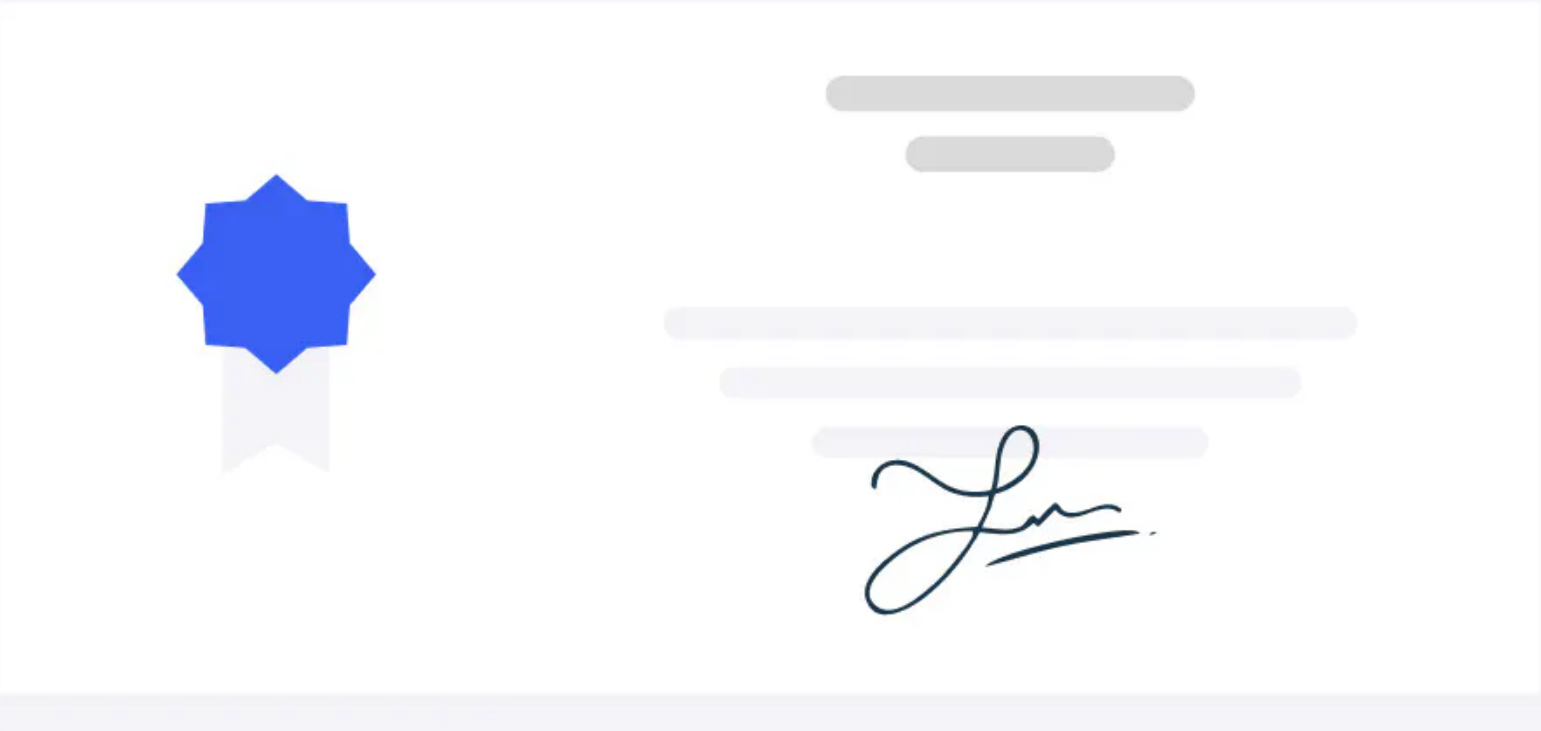
When our team sat together to brainstorm on ideas to make the interview process easier for candidates, we realized quickly that one skill helped us the most when we were preparing for coding interviews: "the ability to map a new problem to an already known problem."

To help candidates with that, we have come up with a list of 27 common coding patterns to help solve coding questions. The idea is, once you are familiar with a pattern, you'll be able to solve dozens of problems with it.

The techniques taught in this course have helped developers land jobs in top companies including Google, Facebook, Amazon, and Microsoft.



Start Reading



Certification on completing the course.

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Course Content

1. Introction — 2 chapters
2. Warmup — 17 chapters
3. Pattern: Two Pointers — 23 chapters
4. Pattern: Fast & Slow Pointers — 15 chapters
5. Pattern: Sliding Window — 21 chapters
6. Pattern: Merge Intervals — 15 chapters
7. Pattern: Cyclic Sort — 17 chapters
8. Pattern: In-place Reversal of a Linked List — 11 chapters
9. Pattern: Stacks — 16 chapters
10. Pattern: Monotonic Stack (*NEW*) — 13 chapters
11. Pattern: Hash Maps — 13 chapters
12. Pattern: Tree Breadth First Search — 19 chapters
13. Pattern: Tree Depth First Search — 15 chapters
14. Pattern: Graphs — 11 chapters
15. Pattern: Island (Matrix Traversal) — 15 chapters
16. Pattern: Two Heaps — 9 chapters
17. Pattern: Subsets — 19 chapters

18. Pattern: Modified Binary Search — 21 chapters

- Introduction to Modified Binary Search Pattern
- Order-agnostic Binary Search (easy)
- Solution: Order-agnostic Binary Search
- Ceiling of a Number (medium)
- Solution: Ceiling of a Number
- Next Letter (medium)
- Solution: Next Letter
- Number Range (medium)
- Solution: Number Range
- Search in a Sorted Infinite Array (medium)
- Solution: Search in a Sorted Infinite Array
- Minimum Difference Element (medium)
- Solution: Minimum Difference Element
- Bitonic Array Maximum (easy)
- Solution: Bitonic Array Maximum
- Problem Challenge 1: Search Bitonic Array (medium)
- Solution: Problem Challenge 1: Search Bitonic Array
- Problem Challenge 2: Search in Rotated Array (medium)
- Solution: Problem Challenge 2: Search in Rotated Array
- Problem Challenge 3: Rotation Count (medium)
- Solution: Problem Challenge 3: Rotation Count

19. Pattern: Bitwise XOR — 9 chapters

- Introduction to Bitwise XOR Pattern
- Single Number (easy)
- Solution: Single Number
- Two Single Numbers (medium)
- Solution: Two Single Numbers
- Complement of Base 10 Number (medium)
- Solution: Complement of Base 10 Number
- Problem Challenge 1: Flip and Invert an Image (hard)
- Solution: Problem Challenge 1: Flip and Invert an Image

20. Pattern: Top 'K' Elements — 29 chapters

- Introduction to Top 'K' Elements Pattern
- Top 'K' Numbers (easy)
- Solution: Top 'K' Numbers
- Kth Smallest Number (easy)
- Solution: Kth Smallest Number
- 'K' Closest Points to the Origin (easy)
- Solution: 'K' Closest Points to the Origin
- Connect Ropes (easy)
- Solution: Connect Ropes
- Top 'K' Frequent Numbers (medium)
- Solution: Top 'K' Frequent Numbers
- Frequency Sort (medium)
- Solution: Frequency Sort
- Kth Largest Number in a Stream (medium)
- Solution: Kth Largest Number in a Stream
- 'K' Closest Numbers (medium)
- Solution: 'K' Closest Numbers
- Maximum Distinct Elements (medium)
- Solution: Maximum Distinct Elements
- Sum of Elements (medium)
- Solution: Sum of Elements
- Rearrange String (hard)
- Solution: Rearrange String
- Problem Challenge 1: Rearrange String K Distance Apart (hard)
- Solution: Problem Challenge 1: Rearrange String K Distance Apart
- Problem Challenge 2: Scheduling Tasks (hard)
- Solution: Problem Challenge 2: Scheduling Tasks
- Problem Challenge 3: Frequency Stack (hard)
- Solution: Problem Challenge 3: Frequency Stack

21. Pattern: K-way Merge — 11 chapters

- Introduction to K-way Merge Pattern
- Merge K Sorted Lists (medium)
- Solution: Merge K Sorted Lists
- Kth Smallest Number in M Sorted Lists (medium)
- Solution: Kth Smallest Number in M Sorted Lists
- Kth Smallest Number in a Sorted Matrix (hard)
- Solution: Kth Smallest Number in a Sorted Matrix
- Smallest Number Range (hard)
- Solution: Smallest Number Range
- Problem Challenge 1: K Pairs with Largest Sums (hard)
- Solution: Problem Challenge 1: K Pairs with Largest Sum

22. Pattern: Greedy Algorithms — 13 chapters

- Introduction to Greedy Algorithm
- Valid Palindrome II (easy)
- Solution: Valid Palindrome II
- Maximum Length of Pair Chain (medium)
- Solution: Maximum Length of Pair Chain
- Minimum Add to Make Parentheses Valid (medium)
- Solution: Minimum Add to Make Parentheses Valid
- Remove Duplicate Letters (medium)
- Solution: Remove Duplicate Letters
- Largest Palindromic Number (Medium)
- Solution: Largest Palindromic Number
- Removing Minimum and Maximum From Array (medium)
- Solution: Removing Minimum and Maximum From Array

23. Pattern: 0/1 Knapsack (Dynamic Programming) — 13 chapters

- Introduction to 0/1 Knapsack Pattern
- 0/1 Knapsack (medium)
- Solution: 0/1 Knapsack
- Equal Subset Sum Partition (medium)
- Solution: Equal Subset Sum Partition
- Subset Sum (medium)
- Solution: Subset Sum
- Minimum Subset Sum Difference (hard)
- Solution: Minimum Subset Sum Difference
- Problem Challenge 1: Count of Subset Sum (hard)
- Solution: Problem Challenge 1: Count of Subset Sum
- Problem Challenge 2: Target Sum (hard)
- Solution: Problem Challenge 2: Target Sum

24. Pattern: Backtracking — 11 chapters

- Introduction to Backtracking Pattern
- Combination Sum (medium)
- Solution: Combination Sum
- Word Search (medium)
- Solution: Word Search
- Sudoku Solver (hard)
- Solution: Sudoku Solver
- Factor Combinations (medium)
- Solution: Factor Combinations
- Split a String Into the Max Number of Unique Substrings (medium)
- Solution: Split a String Into the Max Number of Unique Substrings

25. Pattern: Trie — 11 chapters

- Introduction to Trie
- Implement Trie (Prefix Tree) (medium)
- Solution: Implement Trie (Prefix Tree)
- Index Pairs of a String (easy)
- Solution: Index Pairs of a String
- Design Add and Search Words Data Structure (medium)
- Solution: Design Add and Search Words Data Structure
- Extra Characters in a String (medium)
- Solution: Extra Characters in a String
- Search Suggestions System (medium)
- Solution: Search Suggestions System

26. Pattern: Topological Sort (Graph) — 15 chapters

- Introduction to Topological Sort Pattern
- Topological Sort (medium)
- Solution: Topological Sort
- Tasks Scheduling (medium)
- Solution: Tasks Scheduling
- Tasks Scheduling Order (medium)
- Solution: Tasks Scheduling Order
- All Tasks Scheduling Orders (hard)
- Solution: All Tasks Scheduling Orders
- Alien Dictionary (hard)
- Solution: Alien Dictionary
- Problem Challenge 1: Reconstructing a Sequence (hard)
- Solution: Problem Challenge 1: Reconstructing a Sequence
- Problem Challenge 2: Minimum Height Trees (hard)
- Solution: Problem Challenge 2: Minimum Height Trees

27. Pattern: Union Find (*NEW*) — 9 chapters

- Introduction to Union Find Pattern
- Redundant Connection (medium)
- Solution: Redundant Connection
- Number of Provinces (medium)
- Solution: Number of Provinces
- Is Graph Bipartite? (medium)
- Solution: Is Graph Bipartite?
- Path With Minimum Effort (medium)
- Solution: Path With Minimum Effort

28. Pattern: Ordered Set (*NEW*) — 9 chapters

- Introduction to Ordered Set Pattern
- Merge Similar Items (easy)
- Solution: Merge Similar Items
- 132 Pattern (medium)
- Solution: 132 Pattern
- My Calendar I (medium)
- Solution: My Calendar I
- Longest Continuous Subarray (medium)
- Solution: Longest Continuous Subarray

29. Pattern: Multi-threaded — 7 chapters

- Introduction to Multi-threaded Pattern
- Same Tree (medium)
- Solution: Same Tree
- Invert Binary Tree (medium)
- Solution: Invert Binary Tree
- Binary Search Tree Iterator (medium)
- Solution: Binary Search Tree Iterator

30. Miscellaneous — 2 chapters

- Kth Smallest Number (hard)
- Solution: Kth Smallest Number

31. Revision — 1 chapter

- Coding Patterns: A Cheat Sheet

32. Test Your Knowledge (Easy) — 11 chapters

- Introduction
- Two Sum (easy)
- Solution: Two Sum
- Valid Perfect Square (easy)
- Solution: Valid Perfect Square
- Best Time to Buy and Sell (easy)
- Solution: Best Time to Buy and Sell Stock
- Valid Parentheses (Easy)
- Solution: Valid Parentheses
- Subtree of Another Tree (easy)
- Solution: Subtree of Another Tree

33. Test Your Knowledge (Medium) — 52 chapters

- Daily Temperatures (Medium)
- Solution: Daily Temperatures
- Group Anagrams (Medium)
- Solution: Group Anagrams
- Decode String (Medium)
- Solution: Decode String
- Valid Sudoku (Medium)
- Solution: Valid Sudoku
- Product of Array Except Self (Medium)
- Solution: Product of Array Except Self
- Maximum Product Subarray (Medium)
- Solution: Maximum Product Subarray
- Container With Most Water (Medium)
- Solution: Container With Most Water
- Palindromic Substrings (Medium)
- Solution: Palindromic Substrings
- Remove Nth Node From End of List (Medium)
- Solution: Remove Nth Node From End of List
- Find Minimum in Rotated Sorted Array (Medium)
- Solution: Find Minimum in Rotated Sorted Array
- Pacific Atlantic Water Flow (Medium)
- Solution: Pacific Atlantic Water Flow
- Validate Binary Search Tree (medium)
- Solution: Validate Binary Search Tree
- Construct Binary Tree from Preorder and Inorder Traversal (medium)
- Solution: Construct Binary Tree from Preorder and Inorder Traversal
- Clone Graph (medium)
- Solution: Clone Graph
- House Robber II (medium)
- Solution: House Robber II
- Decode Ways (medium)
- Solution: Decode Ways
- Unique Paths (medium)
- Solution: Unique Paths
- Word Break (medium)
- Solution: Word Break
- Lowest Common Ancestor of a Binary Search Tree (medium)
- Solution: Lowest Common Ancestor of a Binary Search Tree
- Longest Consecutive Sequence (medium)
- Solution: Longest Consecutive Sequence
- Meeting Rooms II (medium)
- Solution: Meeting Rooms II
- Encode and Decode Strings
- Solution: Encode and Decode Strings
- Number of Connected Components in an Undirected Graph
- Solution: Number of Connected Components in an Undirected Graph
- Graph Valid Tree (medium)
- Solution: Graph Valid Tree
- Implement Trie (Prefix Tree) (medium)
- Solution: Implement Trie (Prefix Tree)
- Design Add and Search Words Data Structure (medium)
- Solution: Design Add and Search Words Data Structure

34. Test Your Knowledge (Hard) — 4 chapters

- Longest Valid Parentheses (hard)
- Solution: Longest Valid Parentheses
- Serialize and Deserialize Binary Tree (hard)
- Solution: Serialize and Deserialize Binary Tree

35. Conclusions — 1 chapter

- Where to Go from Here